A Geant4 optical-photon simulation of light collection in a polished plastic scintillator read out by side-coupled silicon photomultipliers for muon trigger detectors

J. Turko[a,*]

[a]*Institution, Address, City, Country*

Abstract

Plastic scintillator plates read out by silicon photomultipliers (SiPMs) are candidates for large-area, low-cost muon trigger detectors in muon tomography. We present a Geant4 (v11.3.2) Monte Carlo model of a polished polystyrene-based (polyvinyltoluene) scintillator plate of 250 mm×250 mm×10 mm read out by 4 mm×4 mm side-coupled SiPMs, reproducing the detector configurations and reflector treatments reported by Rao *et al.* [J. Appl. Phys. **138**, 114504 (2025)]. The optical response is simulated with the Geant4 unified surface model, including scintillation and Čerenkov generation, bulk absorption and wavelength shifting. Photodetection is implemented through the wavelength- dependent photon detection efficiency (PDE) of the SiPM photocathode, and the bare plate is optically coupled to a black holder to represent frustrated total internal reflection. A matrix of four SiPM layouts and four reflector treatments is evaluated at three beam positions (48 runs, 48 000 events), together with a 25 000-event whole-plate scan. The simulation reproduces the measured ordering of light collection, bare $\ll$ TiO_2 $<$ Al $\approx$ glossy, in every layout, and predicts a position contrast below 4 % for the specular reflectors. The wavelength-shifting fibre configuration shifts the mean detected wavelength from 460 nm to 485 nm. The results quantify the role of the reflector in homogenising the response and provide a validated optical model for the design of large-area trigger detectors.

Keywords: Monte Carlo simulation, Geant4, plastic scintillator, silicon photomultiplier, light collection, muon tomography

1. Introduction

Cosmic-ray muons are used to image large, dense objects in applications ranging from volcano and pyramid surveys to the verification of spent nuclear fuel in dry storage casks and to border security [1, 2]. The mean muon energy at sea level is approximately 4 GeV and the flux is about 1 cm^{-2} min^{-1}, with a $\cos^2\theta$ zenith distribution; because muons lose energy slowly, they traverse metre-scale shielding and their scattering and absorption carry information about the material they cross [3]. Large-area plastic scintillator panels read out by photomultipliers are the established technology for the trigger planes of such systems. The transition from photomultiplier tubes (PMTs) to silicon photomultipliers (SiPMs) is motivated by the low cost, low bias voltage, compactness and mechanical robustness of SiPMs, which makes them attractive for modular and mobile detectors [4]. The principal difficulty is the small active area of a SiPM compared with the area of the scintillator plate and its dark count rate: the number of photoelectrons produced by a muon must exceed the thermal noise baseline, which typically requires a reflector treatment that retains and redirects scintillation light towards the photosensors [3].

*Corresponding author
Email address: (J. Turko)

Rao *et al.* [3] have reported an experimental study of 250 mm×250 mm×10 mm polished scintillator plates read out by side-coupled SiPMs, in four configurations (eight, four and two SiPMs, and a single SiPM coupled to a wavelength-shifting (WLS) fibre clamp) and with four surface treatments (bare, white reflector paint, aluminium foil and high-gloss paper). They found that reflective treatments raise the collected light yield by more than an order of magnitude and that the high-gloss paper and aluminium foil perform best, and they proposed the single-SiPM WLS-fibre configuration as a cost-effective design for large-area detectors.

The present work develops an independent Geant4 Monte Carlo model of the same detector and uses it to reproduce the measured ordering of light collection and to expose the photon-transport mechanisms behind it. Optical photon tracking is difficult to model analytically because it is a many-bounce random walk whose outcome depends on the surface finishes. We therefore describe the geometry and optical parameters of every volume and surface, state the physics processes used, and validate the model against the experimental ordering. In addition to the three measurement positions of Ref. [3], a whole-plate scan is performed to quantify the spatial uniformity of the response, which is a design-relevant quantity for a trigger detector.

The article is organised as follows. Section 2 describes the detector model, the optical parameters and the simulation campaign. Section 3 presents the collection efficiency, the photon spectra and the spatial response. Section 4 discusses the physics interpretation and the limitations of the model, and Section 5 concludes.

2. Materials and methods

2.1. Detector model

The active volume is a polished polyvinyltoluene (PVT) scintillator plate of 250 mm×250 mm×10 mm, corresponding to the UPS-92S plate of Ref. [3] and representative of the EJ-200 class [5]. The plate is divided into a 100-point surface grid; its faces are either instrumented with SiPMs or covered by a reflector. Each SiPM is modelled mechanically as a 4 mm×4 mm×0.6 mm block of SiPM epoxy, coupled to the plate through a 0.1 mm layer of optical grease, so that the optical path is plate → grease → SiPM. The sensors are placed as siblings of the plate in the world volume and are clamped so that they do not overhang the plate face.

Four layouts are studied, matching the four configurations of Ref. [3] (Fig. 1):

Config 1 (8 SiPMs): two sensors on each of the four lateral faces, at ±50% of the half-length.

Config 2 (4 SiPMs): one sensor on each face at the Config 1 offsets.

Config 3 (2 SiPMs): both sensors on the same face, at ±50%, reproducing the T1/T2 arrangement of Ref. [3].

Config 4 (1 SiPM + WLS fibre clamp): a single sensor read out through four WLS fibres held in a machined clamp along one narrow face (Sec. 2.3).

The non-instrumented faces receive one of four treatments: no reflector (the plate is held in a black holder), white reflector paint, aluminium foil, or high-gloss paper. The mapping from treatment to optical-surface parameters is described in Sec. 2.2.

2.2. Optical parameters and surfaces

Optical photons are transported with wavelength-dependent properties assigned to every material and surface. The refractive index and bulk attenuation length of the model materials are listed in Table 1; the emission spectrum is tabulated on the same grid. The scintillator emission peaks at 425 nm, and the bulk attenuation length has a maximum of 3.8 m that decreases steeply in the ultraviolet to represent self-absorption. The SiPM photon detection efficiency is tabulated as a function of wavelength, peaking at 0.43 near 425 nm, following the ASD-NUV4S-P-40 device of Ref. [3].

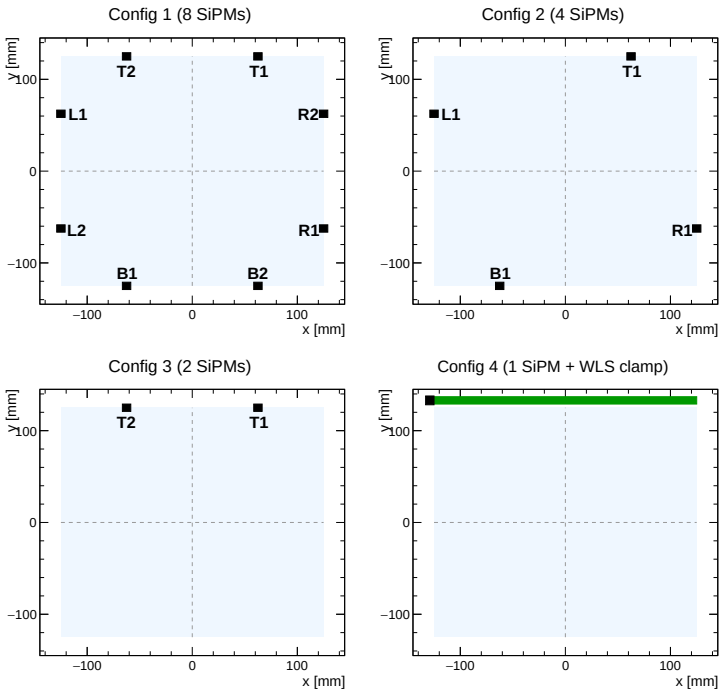


Figure 1: Schematic plan view of the four SiPM layouts on the 250 mm×250 mm plate. Black squares mark the side-mounted SiPMs (drawn oversized; the active area is 4 mm×4 mm). Configs 1–3 use direct coupling; Config 4 uses a single SiPM reading out a WLS fibre clamp along one edge (green bar), with the read-out sensor at its end.

The reflector treatments are implemented as logical surface skins applied to the plate faces, using the Geant4 unified surface model [6, 7]. The parameters are summarised in Table 2. The white paint is modelled as a front-painted Lambertian surface with reflectivity between 0.955 and 0.983; aluminium foil and glossy paper are modelled as back-painted surfaces with a specular spike, i.e. as mirrors, with reflectivities of 0.84–0.91 and 0.90–0.955, respectively. This choice reflects the experimental result that aluminium and glossy paper are the brightest treatments: a purely diffuse model of glossy paper does not reproduce the observation, whereas a slightly lossy mirror does.

The bare configuration requires particular care. A polished plate surrounded by air traps light by total internal reflection (TIR) and would therefore be the most efficient light guide, which contradicts the measurement in which the bare plate gives the weakest signal. In the experiment the plate is mounted in a black 3D-printed holder whose refractive index is close to that of the scintillator; the faces are then in effective optical contact and TIR is frustrated, so that trapped light leaks into the absorbing walls. This is represented in the model by an absorber skin with $R = T = 0$ on the plate faces and a matching skin on the black holder. The same absorber skin is retained on the five non-instrumented faces of Config 4.

Photodetection is delegated to the boundary process rather than drawn in the sensitive detector. The grease–SiPM boundary is modelled as a `dielectric_metal` surface whose efficiency equals the SiPM PDE and whose reflectivity is zero. When a photon reaches this boundary, the boundary process performs the absorption draw, assigns the status `Detection` and invokes the sensitive detector; the sensitive detector records one photoelectron and terminates the photon. The Geant4 option `SetBoundaryInvokeSD(true)` is enabled for this purpose. The plate–grease boundaries are transparent Fresnel surfaces, so that the coupling chain is governed by the refractive-index mismatch alone.

2.3. Config 4: WLS fibre clamp

Config 4 follows the single-SiPM, WLS-fibre read-out of Ref. [3]. Four Y-11(200) fibres of 1.0 mm diameter run along one narrow face over its full 250 mm length at a 1.5 mm pitch and converge, over the last

Table 1: Refractive index n and bulk attenuation length L_{abs} of the model materials. WLS = wavelength-shifting fibre.

Material	Role	n	L_{abs}
PVT scintillator	active volume	1.58	3.8 m (peak)
Optical grease	plate–SiPM coupling	1.46	—
SiPM epoxy	photosensor body	1.55	—
Black holder	light-tight enclosure	1.55	1 µm
WLS fibre core	Y-11(200) core	1.59	5 m
WLS fibre cladding	fibre cladding	1.49	20 m
Fibre-clamp body	black clamp	1.55	1 µm

Table 2: Optical-surface model for the four reflector treatments. R is the tabulated reflectivity over 360 nm to 650 nm. All surfaces use the unified model.

Treatment	Geant4 type	Geant4 finish	Reflectivity R
Bare / black holder	`dielectric_dielectric`	`polished` (absorber)	$T = R = 0$
White paint (TiO_2)	`dielectric_dielectric`	`groundfrontpainted`	0.955–0.983, Lambertian
Aluminium foil	`dielectric_dielectric`	`polishedbackpainted`	0.84–0.91, specular
Glossy paper	`dielectric_dielectric`	`polishedbackpainted`	0.90–0.955, specular

15 mm, to a touching 1.0 mm bundle that presents all four ends to a single SiPM. The far ends are closed by aluminium mirror caps. The fibres are held in a black clamp with a 1.1 mm channel per fibre, separated from the plate by a 0.10 mm grease film; each fibre protrudes 0.05 mm into the film so that the grease remains continuous against the scintillator. The 0.05 mm air annulus between the 1.0 mm fibre and its channel is retained because the modelled cladding index ($n = 1.49$) exceeds the grease index ($n = 1.46$) only slightly; a snug fit against the clamp would frustrate core–cladding TIR. The fibres are built as tessellated swept cylinders and placed as world siblings of the plate. The Geant4 WLS process is enabled automatically for this geometry.

2.4. Physics processes and Monte Carlo implementation

The simulation uses Geant4 v11.3.2 [8, 9] with a modular physics list built from the standard electromagnetic, hadronic elastic and inelastic, thermal-neutron, ion and radioactive-decay constructors. Optical photons are produced by the scintillation and Čerenkov processes and transported with bulk absorption, Rayleigh scattering and boundary processes enabled; Mie scattering is disabled. Scintillation is described by a single fast component with a decay time of 2.1 ns and a light yield of 1.0×10^4 photons per MeV; the energy dependence of the light yield is modelled with Birks' law with a constant of $0.126\,\mathrm{mm\,MeV^{-1}}$ [10, 11]. The boundary process requires a refractive index on both sides of an interface; every material in the model therefore carries an explicit refractive-index table, and a missing index would silently terminate the photon. The wavelength-shifting process is enabled only for the Config 4 geometry and is inactive for all other layouts.

The implementation of the optical surfaces was verified against the Geant4 source. Reading the boundary process, a logical border surface between two volumes takes precedence over a logical skin surface at the same boundary, and a boundary with no surface object falls back to lossless TIR. These two properties are used deliberately: the grease and SiPM patches punch holes in the reflector skin, and the bare plate is given an absorber skin rather than a reflective one.

2.5. Simulation campaign and efficiency definition

The primary source is a vertical 4 GeV μ^- pencil beam, generated with the Geant4 General Particle Source (GPS), with a square cross-section of 50 mm×50 mm matching the active area of the muon coincidence detector (MCD) of Ref. [3]. Three irradiation positions are used, each illuminating a 50 mm×50 mm patch lying entirely on the plate: the centre, an edge mid-point, and a corner. The edge and corner patches

Table 3: Simulated collection efficiency (all SiPMs and all irradiation positions combined).

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	1.9×10^{-4}	9.7×10^{-3}	2.64×10^{-2}	2.99×10^{-2}
Config 2 (4 SiPMs)	9.6×10^{-5}	4.9×10^{-3}	1.38×10^{-2}	1.58×10^{-2}
Config 3 (2 SiPMs)	2.4×10^{-5}	1.30×10^{-3}	7.20×10^{-3}	8.18×10^{-3}
Config 4 (1 SiPM, WLS)	6.9×10^{-6}	2.2×10^{-4}	1.35×10^{-3}	1.47×10^{-3}

lie on the face opposite the instrumented face of the single-sided layouts. The full matrix comprises four layouts, four reflector treatments and three positions, i.e. 48 runs of 1000 events each. A second, whole-plate campaign replaces the pencil beam with a 250 mm×250 mm beam so that the muon entry point is sampled uniformly over the plate; 25 000 events are generated for each (layout, reflector) pair. All runs are executed with eight threads and without biasing.

For a set of SiPMs $\mathcal{S}$ the collection efficiency is defined as

$$\varepsilon = \frac{\sum_{e} \sum_{s \in \mathcal{S}} N_{\mathrm{pe}}^{(e,s)}}{\sum_{e} N_{\gamma}^{(e)}}, \tag{1}$$

where $N_{\mathrm{pe}}^{(e,s)}$ is the number of photoelectrons detected in sensor s in event e and $N_{\gamma}^{(e)}$ is the total number of optical photons generated in the plate during that event (scintillation and Čerenkov). The numerator and denominator are accumulated per event, so that the efficiency is a ratio of sums and is independent of the number of sensors.

A dedicated diagnostic record accumulates, per event, the number of generated, detected, escaped and terminated optical photons, the per-event produced-photon count and the primary muon's first entry position in the plate. At the end of each event the sensitive detector writes one row per (event, sensor), including sensors with zero detections; the sample is therefore unbiased in the muon interaction position. Each row carries the entry position, the produced-photon count, the detected count, the first arrival time and the mean detected wavelength. Position-resolved efficiencies are reconstructed offline from this record in 5 mm (matrix) or 10 mm (whole plate) cells.

3. Results

3.1. Collection efficiency

Figure 2 shows the simulated collection efficiency, Eq. (1), summed over all SiPMs and over the three irradiation positions, for every (layout, reflector) pair; the values are listed in Table 3. In every layout the ordering is

$$\text{bare} \ll TiO_2 < \text{Al} \approx \text{glossy}, \tag{2}$$

which reproduces the experimental result of Ref. [3]. The bare plate is suppressed by two orders of magnitude or more relative to the reflective treatments, as expected for a surface whose faces are in effective optical contact with an absorber. Aluminium and glossy paper are the most efficient treatments and are indistinguishable within the statistical precision of the sample. The efficiency increases with the number of sensors: Config 1 (eight SiPMs) is the most efficient, followed by Config 2 (four) and Config 3 (two). The single-SiPM WLS-fibre layout (Config 4) collects the least light, since only the fraction of photons that reach the fibre face is converted and guided to the sensor.

3.2. Photon yield and photoelectron spectra

Figure 3(a) shows the per-event optical-photon yield (the denominator of Eq. (1)) for the four treatments. The mean yield is essentially independent of the reflector, at approximately 19 000 photons per event,

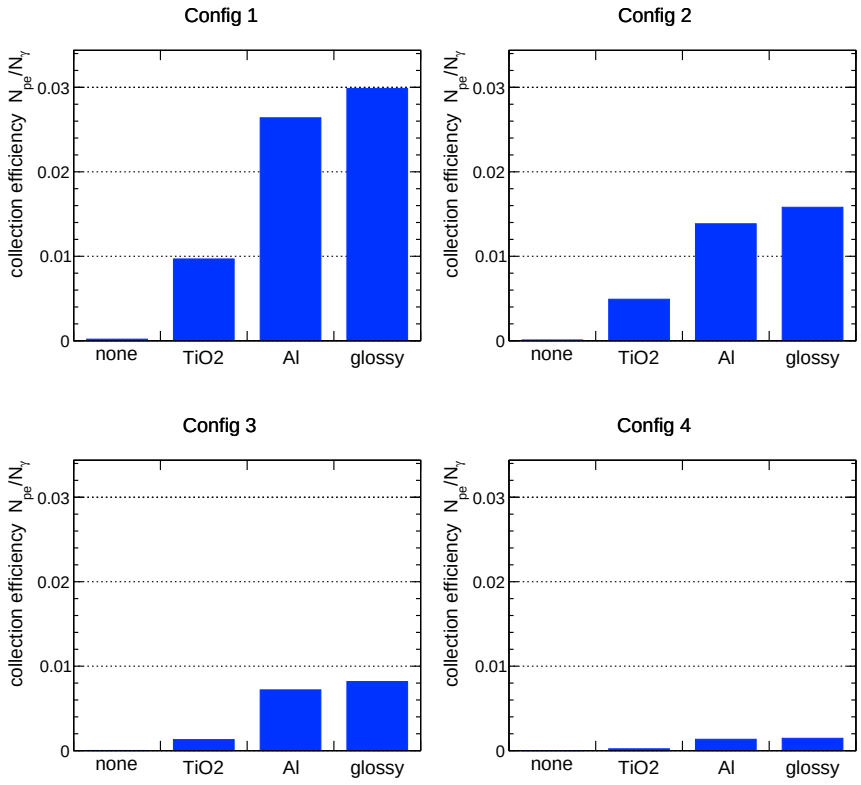


Figure 2: Simulated collection efficiency, Eq. (1), for the four reflector treatments and four SiPM layouts. The ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy is reproduced in every layout.

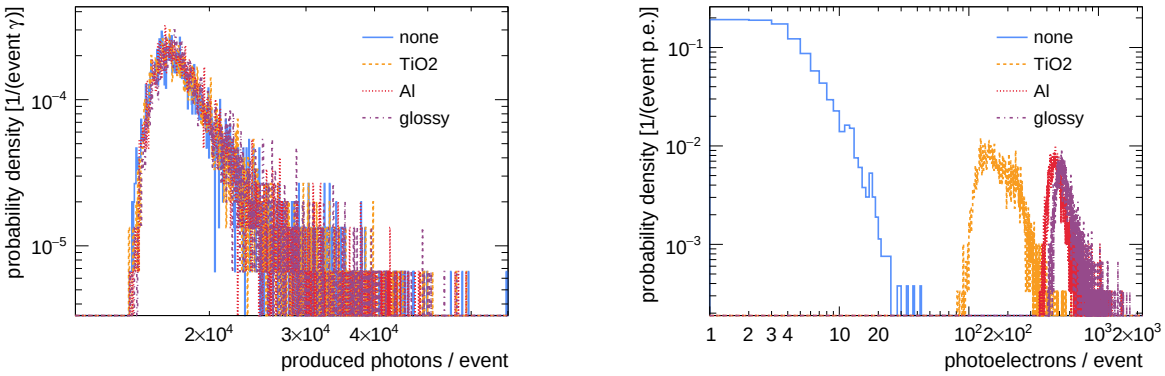
as required: the reflector changes the transport of photons, not their production. The residual spread reflects the event-to-event variation of the muon path length for a vertical beam of fixed cross-section. Figure 3(b) shows the corresponding photoelectron spectra, in which the reflector has a decisive effect. The bare configuration peaks near zero, with a mean of 3.7 photoelectrons per event for Config 1, whereas glossy paper gives a mean of 586 photoelectrons per event. The transition from a near-zero-peaked to a broad distribution is the direct signature of efficient light collection. With approximately 19 000 photons produced and $\approx$ 586 photoelectrons detected, the collection efficiency is $\approx 3\,\%$, consistent with the SiPM PDE of ≈ 0.43 and the geometric acceptance of a side-coupled sensor.

3.3. Detected wavelength

Figure 4 shows the wavelength distribution of the detected photoelectrons for each treatment and layout. Configs 1–3, which couple the SiPMs directly to the plate, detect the same blue scintillation spectrum; the mean wavelength varies only from 459.8 nm (glossy, Config 1) to 460.1 nm (Config 3). The spectrum is red-shifted with respect to the 425 nm emission peak because the short-wavelength part of the emission overlaps the absorption band and is attenuated over the multi-bounce path. Config 4 is qualitatively different: the WLS fibres absorb the blue light and re-emit it at their own emission band, shifting the mean detected wavelength to 485.2 nm (glossy), a shift of approximately 25 nm. The same shift appears for every treatment, confirming that it originates in the fibre wavelength shifter rather than in the reflector.

3.4. Position dependence and light sharing

The three irradiation patches have equal area and lie entirely on the plate, so their produced-photon denominators are equal and their efficiencies are directly comparable. Figure 5 presents the efficiency at the



(a) Produced optical photons per event.

(b) Detected photoelectrons per event.

Figure 3: Per-event distributions for Config 1 (unit-area probability density; logarithmic axes): (a) the production yield is reflector-independent, while (b) the detected photoelectron count depends strongly on the reflector.

Table 4: Position contrast $\kappa = \varepsilon_{\max}/\varepsilon_{\min}$ (Eq. (3)) across the three irradiation patches.

Treatment	Config 1	Config 2	Config 3	Config 4
Bare	2.22	1.98	3.33	3.73
TiO_2	1.74	1.72	1.81	2.09
Al	1.03	1.03	1.01	1.03
Glossy	1.03	1.03	1.00	1.03

centre, edge and corner for each layout. The reflective treatments homogenise the response: for aluminium and glossy paper the efficiency varies by less than 4 % across the three patches in every layout. The diffuse TiO_2 paint is less uniform, and the bare plate shows the strongest position dependence. This behaviour is quantified by the position contrast

$$\kappa = \frac{\varepsilon_{\max}}{\varepsilon_{\min}}, \tag{3}$$

the ratio of the largest to the smallest patch-averaged efficiency, listed in Table 4. The mirror-like treatments hold $\kappa \leq 1.03$, whereas κ reaches 1.7–2.1 for TiO_2 and up to 3.7 for the bare plate. The result is that a closed reflective cavity retains little memory of the muon entry point after a few bounces; the sensor coverage, rather than the irradiation position, then sets the collection efficiency.

Figure 6 resolves the directional response for Config 1 by showing the share of the collected light detected by each of the eight sensors. For the bare plate at the corner patch, the two sensors nearest the beam collect 91 % of all photoelectrons, approximately 3.6 times the uniform share each, while two of the remaining sensors record nothing. A reflective treatment removes this structure: for glossy paper the same two sensors reach only about 1.1 times the uniform share. The total efficiency is nearly the same for all three patches, so the cavity keeps the amount of collected light constant while redistributing it among the sensors.

3.5. Position-resolved efficiency maps

Figure 7 shows the collection efficiency as a function of the muon entry point for Config 1, in 5 mm cells, one panel per treatment. Only the three irradiated patches are populated. Within a patch the reflective treatments are nearly featureless, consistent with the uniform integrated response found above, whereas the bare plate and, to a lesser degree, the TiO_2 coating show a clear gradient across the field, brightest on the side nearest a sensor.

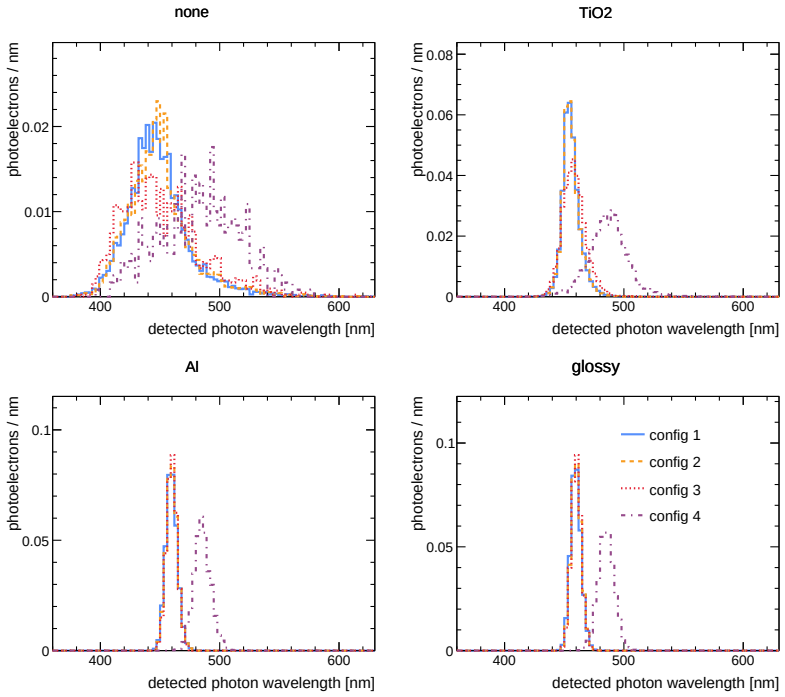


Figure 4: Detected-photon wavelength distribution, weighted by photoelectron count, for each reflector (panel) and each layout (curves). Configs 1–3 follow the blue scintillation spectrum near 460 nm, whereas the WLS-fibre read-out of Config 4 is shifted to approximately 485 nm.

3.6. Whole-plate uniformity

The three-patch scan resolves the centre, edge and corner but not the whole plate. A second campaign irradiates the full 250 mm×250 mm surface uniformly with 25 000 events per (layout, treatment) pair. Figure 8 shows the resulting maps for Config 1. The bare plate exhibits a pronounced spatial structure: the efficiency is lowest near the centre and rises towards the instrumented edges, where photons are created close enough to a sensor-bearing face to be detected before the frustrated-TIR loss into the black holder can absorb them. The reflective panels reproduce the ordering of the matrix scan with the position dependence progressively removed; for aluminium and glossy paper the centre and the outskirts agree to better than 1 %.

To quantify the response over the whole plate we bin the events on a 5×5 grid of 50 mm cells and form the contrast

$$\kappa = \frac{\max_{\text{cell}} \varepsilon}{\min_{\text{cell}} \varepsilon}, \tag{4}$$

so that $\kappa = 1$ denotes a uniform response and $\kappa - 1$ the fractional spread over the plate. Unlike a purely radial average, this metric retains the azimuthal information that distinguishes the single-sided layouts. The values are listed in Table 5. The treatments are ranked identically and monotonically in every layout,

$$\text{bare} \gg \text{TiO}_2 > \text{Al} \gtrsim \text{glossy}, \tag{5}$$

in agreement with the three-patch ordering. The whole-plate contrast is numerically larger than the three-patch value because the full plate includes the extreme corners that the three patches average over: κ reaches 15.7 for the bare plate in Config 2 and 138 in Config 4. The mirror-like treatments keep $\kappa < 1.6$ in every layout. The single-sided layouts are the discriminating case: for Config 4, whose WLS fibre runs the full length of one edge, the near–far asymmetry across the plate is the strongest of the study, giving the largest whole-plate contrast, whereas the four-sided layouts (1 and 2) remain nearly uniform.

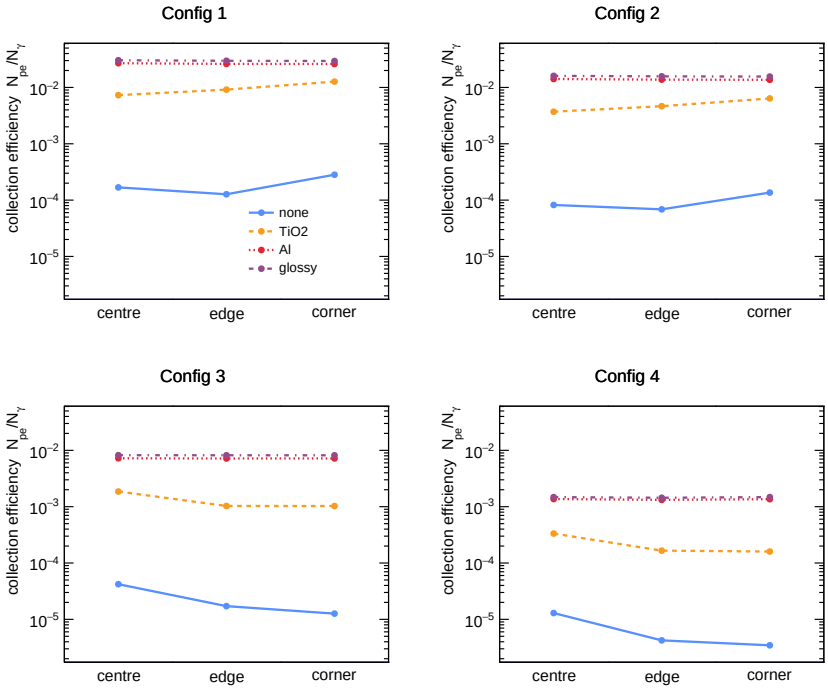


Figure 5: Collection efficiency at the centre, edge and corner patches, one panel per layout, for the four treatments (logarithmic axis). The reflective treatments are nearly flat across the three patches, whereas the bare plate is both the weakest and the most position-dependent.

Table 5: Whole-plate contrast $\kappa = \max_{\text{cell}} \varepsilon / \min_{\text{cell}} \varepsilon$ over 50 mm cells, from 25 000 events per (layout, treatment) pair. A value of unity denotes a uniform response.

Treatment	Config 1	Config 2	Config 3	Config 4
Bare	9.5	15.7	67	138
TiO_2	1.90	2.47	8.15	16.9
Al	1.08	1.13	1.29	1.51
Glossy	1.06	1.11	1.24	1.48

4. Discussion

4.1. Origin of the reflector ordering

The simulation reproduces the measured ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy and explains it in terms of the photon-transport regime rather than in terms of a fitted collection parameter. For the bare plate the non-instrumented faces are in effective optical contact with a black holder whose refractive index is close to that of the scintillator, so total internal reflection is frustrated and light leaks into the walls; the result is the weakest response of the study and a pronounced edge-read-out structure, in which photons created near an instrumented face are collected before the loss to the holder can act. A reflective treatment reverses this: photons that reach an outer face are returned to the plate and are eventually collected, which raises the efficiency by one to more than two orders of magnitude and, more importantly, makes the response almost independent of the muon entry point.

The two classes of reflector differ in how they mix the light. The white paint returns light through a purely Lambertian (diffuse) channel with no total internal reflection, so the reflected light is randomised in direction and the position dependence is reduced but not erased. Aluminium foil and glossy paper are

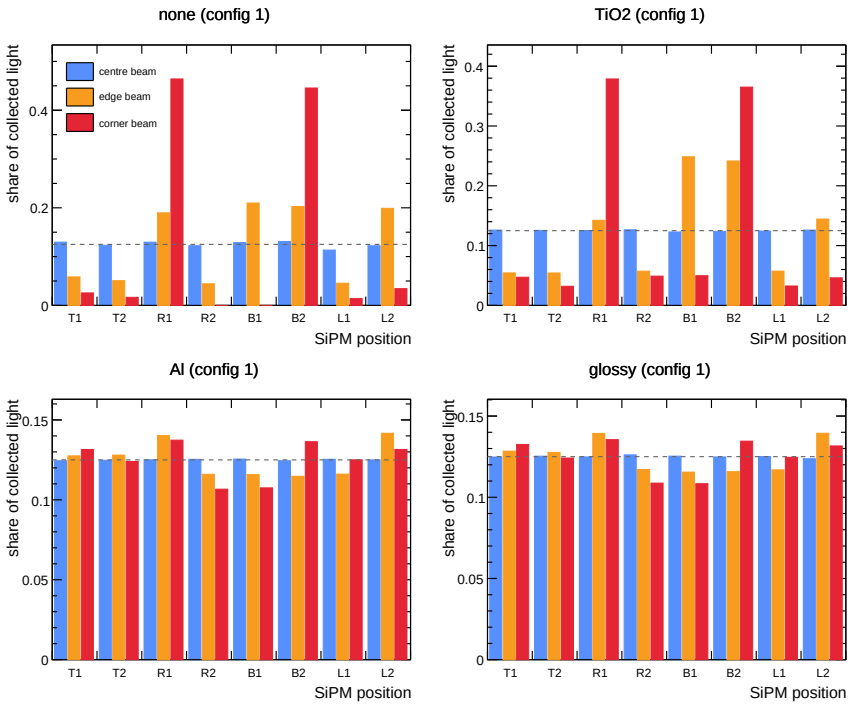


Figure 6: Direction-resolved light sharing for Config 1, one panel per treatment. Each group of bars is one sensor; the three colours are the irradiation positions; the dashed line marks the uniform expectation 1/8. For the bare plate the light remains concentrated on the two nearest sensors, whereas the reflective treatments distribute it almost uniformly.

modelled as lossy specular mirrors, and a specular cavity mixes the light far more effectively: the position contrast stays below 4 % in every layout, the whole-plate response is flat to better than 1 %, and the per-sensor light sharing becomes nearly uniform. The experimental finding that aluminium and glossy paper perform equally well, and better than the diffuse white paint, therefore distinguishes a mirror-like treatment from a diffusive one and is reproduced only when glossy paper is modelled as a slightly lossier mirror rather than as a diffuser. The ordering and the homogenisation are robust: they hold for every sensor layout, for both the three-patch and the whole-plate campaigns, and the whole-plate contrast separates the treatments in the same monotone order as the three-patch contrast.

4.2. Comparison with the experimental study

The model reproduces the qualitative features reported in Ref. [3]: the strong suppression of the bare plate, the effective equivalence of aluminium and glossy paper as the brightest treatments, the blue detected spectrum near 460 nm for direct side coupling, and the green-shifted spectrum produced by the wavelength-shifting fibre read-out. It also quantifies the spatial uniformity that the experiment samples only at three positions, showing that the reflector, and not the sensor layout, governs whether the entry point is remembered.

One quantitative comparison must be made with care. In our model the single-SiPM fibre layout (Config 4) collects less light than the two-SiPM direct layout (Config 3), whereas Ref. [3] reports the opposite ordering. The two configurations are, however, not directly comparable, as Ref. [3] itself notes: they differ in sensor count, gain and input capacitance, and the reported Config 4 quantity is a pulse height over a fibre-augmented effective area rather than a photoelectron count. We therefore regard the absolute cross-configuration scale as model-dependent and do not claim a discrepancy; the physics of the fibre read-out, namely the wavelength-shifted and strongly position-dependent character of the collected light, is reproduced.

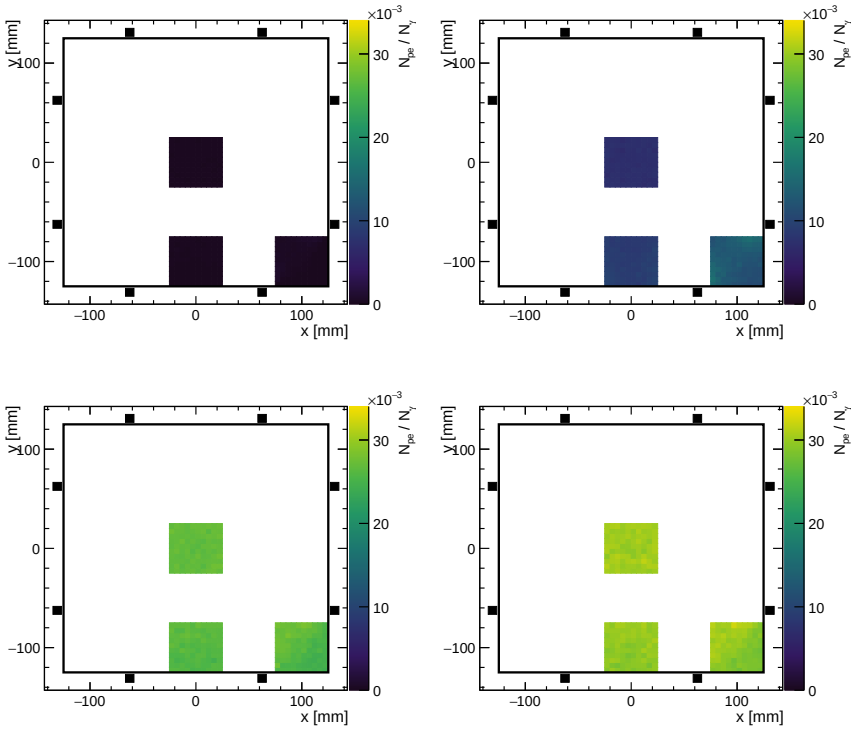


Figure 7: Collection efficiency map for Config 1, one panel per treatment (5 mm cells). Black squares mark the SiPM positions. The reflective treatments are nearly uniform within each irradiated patch.

4.3. Limitations of the model

The absolute efficiencies are fixed by the assumptions of the model—the photocathode efficiency curve, the grease coupling, and the absorbed fraction of the tabulated reflectivities—and should not be interpreted as predictions of absolute light yield. The robust, model-independent results are the ordering of the treatments, the splitting into an effectively absorbing bare case and a reflective family, and the spatial homogenisation produced by the specular reflectors.

Two further limitations are noted. First, the diffuse white-paint configuration shows a reversed position trend in the simulation, corner > edge > centre, whereas Ref. [3] reports no such effect; since the front-painted surface is purely Lambertian, the corner geometry can couple diffusely scattered light into the edge sensors more efficiently, but this prediction should be confirmed with independent simulation settings before being interpreted physically. Second, during the production runs a rare intermittent stall was observed in the multi-threaded optical tracking, in which a single worker thread entered a tight loop while the others waited; the affected run was discarded and regenerated, and all results reported here are drawn from complete sets of files.

5. Conclusions

A Geant4 optical-photon Monte Carlo model of a 250 mm×250 mm×10 mm polished plastic scintillator read out by side-coupled SiPMs has been developed and applied to a matrix of four sensor layouts and four reflector treatments at three beam positions, together with a whole-plate scan. The model uses the Geant4 unified surface model, with photodetection implemented through the wavelength-dependent photocathode efficiency at the grease–SiPM boundary, and the bare plate coupled to a black holder that frustrates total internal reflection.

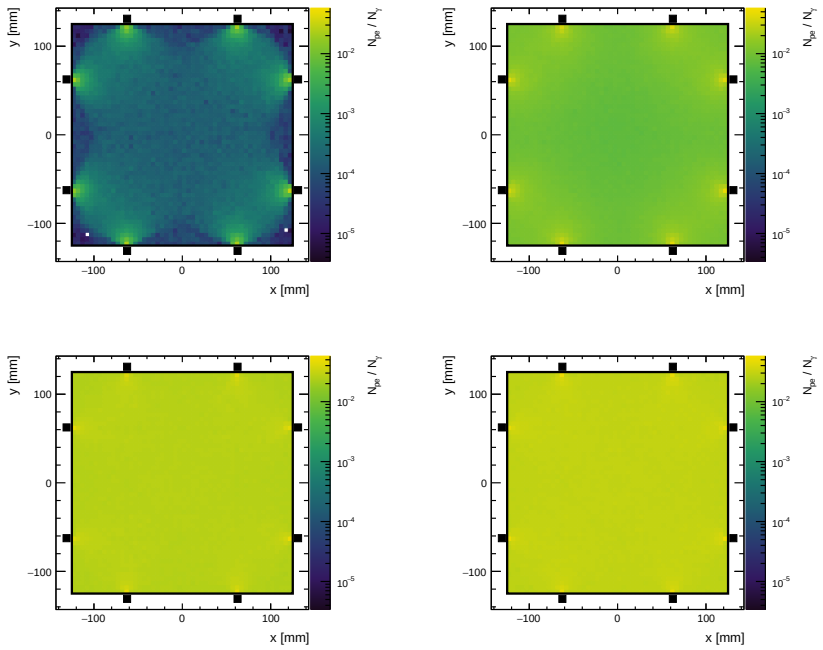


Figure 8: Whole-plate collection-efficiency maps for Config 1 from 25 000 events per panel (10 mm cells, logarithmic colour scale). The bare plate is dimmer at the centre and brighter towards the instrumented edges; the reflective treatments are brighter and nearly flat.

The simulation reproduces the measured ordering of light collection, bare $\ll$ TiO_2 $<$ Al $\approx$ glossy, in every layout, and identifies frustrated total internal reflection into the holder, rather than the intrinsic light yield, as the cause of the weak bare response. The specular reflectors homogenise the response far more effectively than the diffuse paint, holding the position contrast below 4 % and the whole-plate contrast below 1.6 while the bare plate varies by more than an order of magnitude over the plate. The wavelength-shifting fibre read-out shifts the mean detected wavelength from 460 nm to 485 nm, confirming the operation of the shifter. The validated model provides a quantitative basis for optimising the reflector and sensor coverage of large-area plastic-scintillator muon trigger detectors.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation source code, macros and the data generated in this work are available from the author on reasonable request.

Acknowledgements

The author thanks the authors of Ref. [3] for the experimental study that motivated this work. [Funding to be added.]

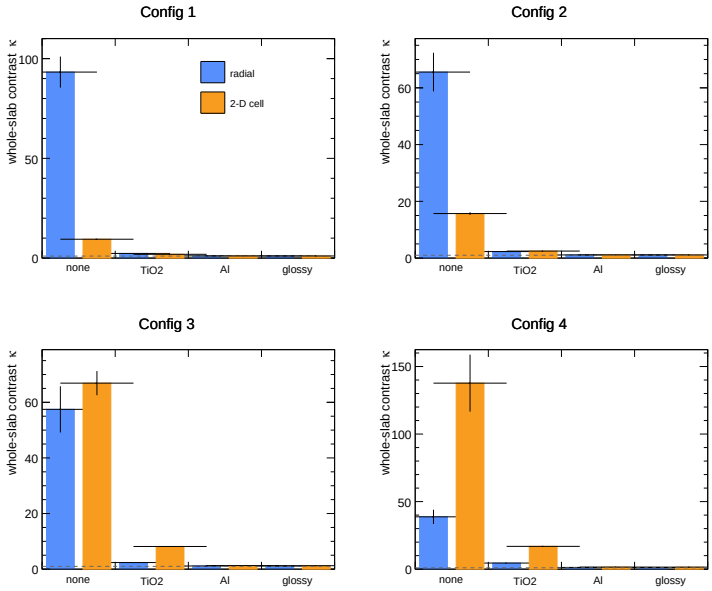


Figure 9: Whole-plate contrast κ (Eq. (4)) from 50 mm cells, one panel per layout, four treatments per panel (dashed line: $\kappa = 1$). Both the ordering and the magnitude separate the absorbing bare plate from the reflective treatments in every layout.

References

[1] K. N. Borozdin, G. E. Hogan, C. Morris, W. C. Priedhorsky, A. Saunders, L. J. Schultz, M. E. Teasdale, Radiographic imaging with cosmic-ray muons, Nature 422 (2003) 277. doi:10.1038/422277a.

[2] L. Bonechi, R. D'Alessandro, A. Giammanco, Atmospheric muons as an imaging tool, Reviews in Physics 5 (2020) 100038. doi:10.1016/j.revip.2020.100038.

[3] G. Rao, A. Bieberle, U. Hampel, M. Wagner, Design studies on large muon trigger detectors using silicon photomultiplier devices, Journal of Applied Physics 138 (11) (2025) 114504. doi:10.1063/5.0288270.

[4] S. Gundacker, A. Heering, The silicon photomultiplier: fundamentals and applications of a modern solid-state photon detector, Physics in Medicine & Biology 65 (17) (2020) 17TR01. doi:10.1088/1361-6560/ab7b2d.

[5] Eljen Technology, EJ-200, EJ-204, EJ-208, EJ-212 general purpose plastic scintillators – data sheet, Vendor datasheet, https://eljentechnology.com/images/products/data_sheets/EJ-200_EJ-204_EJ-208_EJ-212.pdf (2024).

[6] A. Levin, C. Moisan, A more physical approach to model the surface treatment of scintillation counters and its implementation into DETECT, in: 1996 IEEE Nuclear Science Symposium Conference Record, Vol. 2, 1996, pp. 702–706. doi:10.1109/NSSMIC.1996.591410.

[7] J. Nilsson, V. Cuplov, M. Isaksson, Identifying key surface parameters for optical photon transport in GEANT4/GATE simulations, Applied Radiation and Isotopes 103 (2015) 15–24. doi:10.1016/j.apradiso.2015.04.017.

[8] S. Agostinelli, J. Allison, K. Amako, J. Apostolakis, H. Araujo, et al., Geant4—a simulation toolkit, Nuclear Instruments and Methods in Physics Research A 506 (3) (2003) 250–303. doi:10.1016/S0168-9002(03)01368-8.

[9] J. Allison, K. Amako, J. Apostolakis, P. Arce, M. Asai, et al., Recent developments in Geant4, Nuclear Instruments and Methods in Physics Research A 835 (2016) 186–225. doi:10.1016/j.nima.2016.06.125.

[10] J. B. Birks, Scintillations from organic crystals: Specific fluorescence and relative response to different radiations, Proceedings of the Physical Society. Section A 64 (10) (1951) 874–877. doi:10.1088/0370-1298/64/10/303.

[11] R. L. Craun, D. L. Smith, Analysis of response data for several organic scintillators, Nuclear Instruments and Methods 80 (2) (1970) 239–244. doi:10.1016/0029-554X(70)90768-8.